

# AGNIBH DASGUPTA

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## Education

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CURRENT    PhD, Computing & Information Science, **University of Nebraska, Omaha**  
2018 - 2020    Master of Science, Computer Science, **Utah State University**  
2013 - 2017    Bachelor of Technology, Computer Science, **West Bengal University of Technology**

## Publications

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- 2025    **Watermarking Language Models through Language Models (IEEE TAI)** [↗](#) (*in press*)
- Built a prompt-based LLM watermarking framework without modifying model weights or data
  - Evaluated watermark generation and detection using instruction-tuned LLMs
- 2025    **MAE-SAM2: Mask Autoencoder-Enhanced SAM2 for Clinical Retinal Vascular Leakage Segmentation (AAO)** [↗](#)
- Trained a classifier on patches of large optic scans to accurately detect areas of leakage
- 2024    **Enhanced Image Watermarking Through Cross-Attention and Noise-Invariant Domain Learning (Imaging Science: Computer Vision, Image and Signal Processing Pattern Recognition)** [↗](#)
- Improved robustness and scalability of image watermarking under heavy compound noise
- 2024    **Leveraging Artificial Intelligence (AI) to Enhance CS Instruction (FIE)** [↗](#)
- Utilized multi-modality in deep learning models for mood detection in K-12 classroom videos
- 2023    **Robust Image Watermarking based on Cross-Attention and Invariant Domain Learning (CSCI)** [↗](#)
- Designed a robust image watermarking scheme using cross-attention (ViT)
  - Trained a self-supervised invariant domain to learn semantic features
- 2022    **Perspective transformation layer (CSCI)** [↗](#)
- Proposed a lightweight layer that learns 2D projections of 3D views for multiple viewpoints

## Present and Past Research

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- PRESENT    **Invariant Latent Features in Large Language Models**
- Studying model-specific invariant features for robust model attribution
- PRESENT    **Adversarial Auto Augmenter for Realistic Image Augmentations**
- Learning modular auto-augmentations for robust invariant latent representation
- 2022    **Sentiment Analysis on Twitter data on Electric Vehicles**
- Analyzed sentiment with lexical and Word2Vec-LSTM approaches on scraped twitter data
- 2020    **Deep Q Learning applied to stock trading** [↗](#)
- Used a DQN agent to optimize stock trading on the S&P500 index using Google Trends data

## Reviewer Service

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2022 - 2025    ACCV, Scientific Reports - Nature, Springer Nature

## Teaching Experience

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2023 - 2024    **Graduate teaching assistant** at University of Nebraska at Omaha  
2022 - 2023    **Intern supervisor** at University of Nebraska at Omaha  
2018 - 2020    **Graduate teaching assistant** at Utah State University

## Technical Skills

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|-----------------------------|--------------------------------------------------|
| PROGRAMMING & SYSTEMS:      | Python, Bash, Linux, MATLAB, Git, JAX, CUDA      |
| DEEP LEARNING FRAMEWORKS:   | PyTorch, TensorFlow, Transformers, LLMs, NLP     |
| DISTRIBUTED & OPTIMIZATION: | DeepSpeed, PyTorch DDP, tf.distribute, Slurm/HPC |